Report on insurance cost analysis

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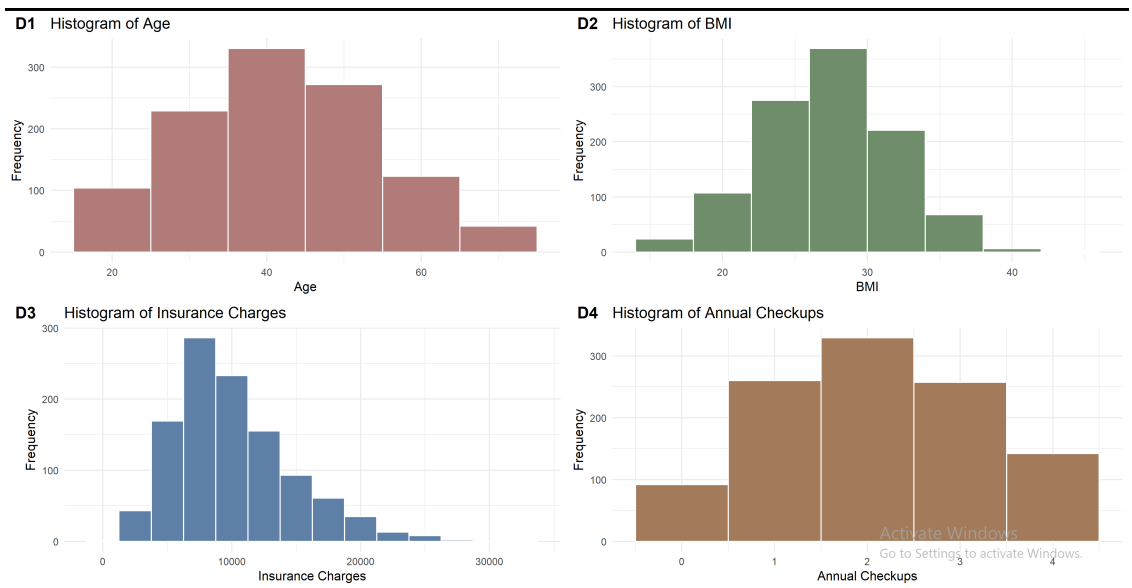
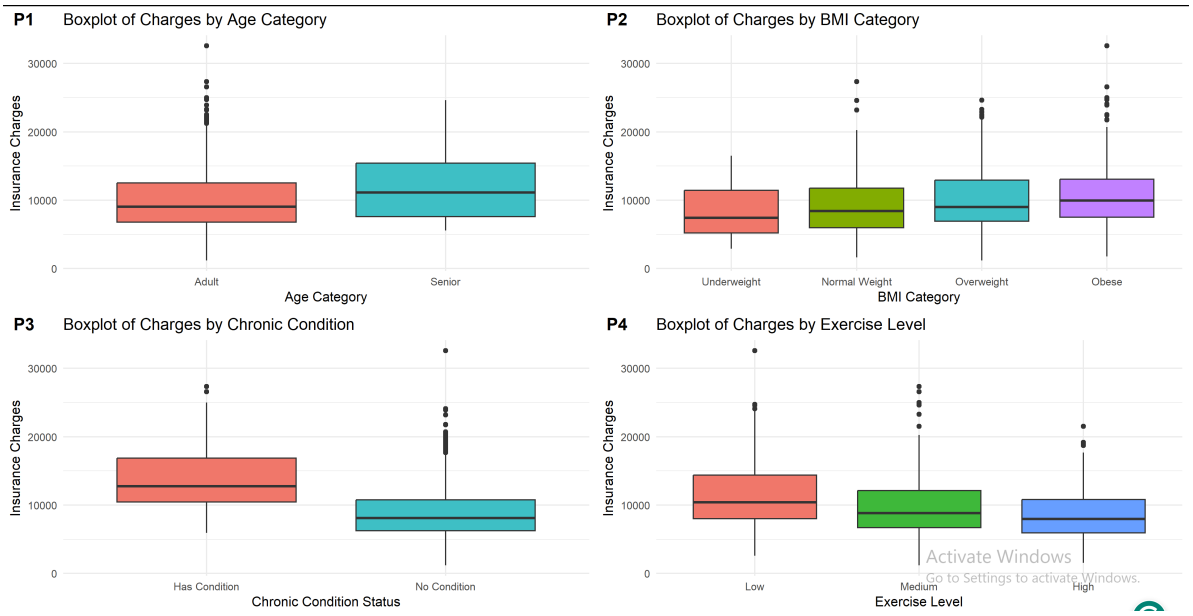
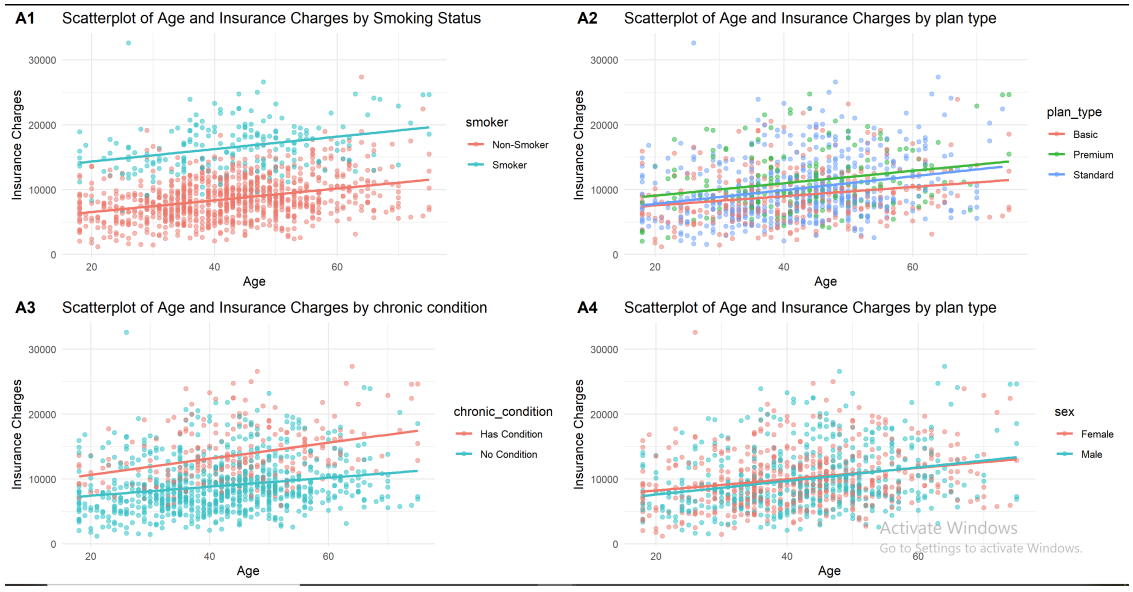
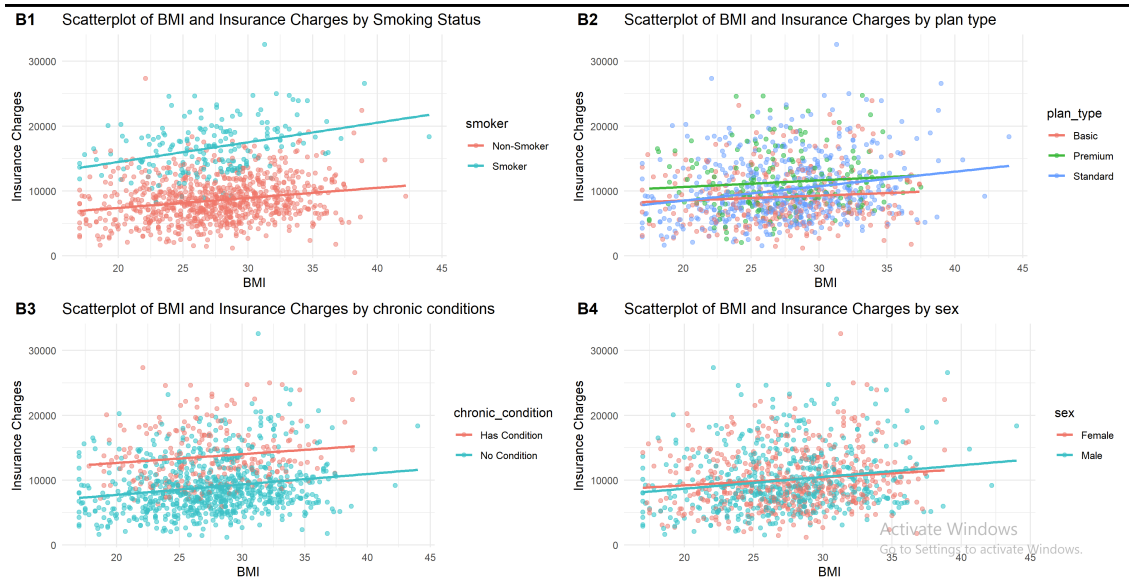
Introduction

variable	n	mean	sd	median	min	max
age	1,031	42.420	12.380	42.00	18.00	75.00
bmi	1,031	27.474	4.606	27.60	17.00	44.00
charges	1,031	10,084.302	4,606.093	9,122.72	1,203.51	32,559.28




P1 Boxplot of Charges by Age Category
Insurance Charges
Adult
Senior
Age Category
P2 Boxplot of Charges by BMI Category
Underweight
Normal Weight
Overweight
Obese
BMI Category
P3 Boxplot of Charges by Chronic Condition
Has Condition
No Condition
Chronic Condition Status
P4 Boxplot of Charges by Exercise Level
Low
Medium
High
Exercise Level
Activate Windows
Go to Settings to activate Windows.


A1 Scatterplot of Age and Insurance Charges by Smoking Status
smoker
Non-Smoker
Smoker
A2 Scatterplot of Age and Insurance Charges by plan type
plan_type
Basic
Premium
Standard
A3 Scatterplot of Age and Insurance Charges by chronic condition
chronic_condition
Has Condition
No Condition
A4 Scatterplot of Age and Insurance Charges by plan type
sex
Female
Male
Insurance Charges
Age
Activate Windows
Go to Settings to activate Windows.



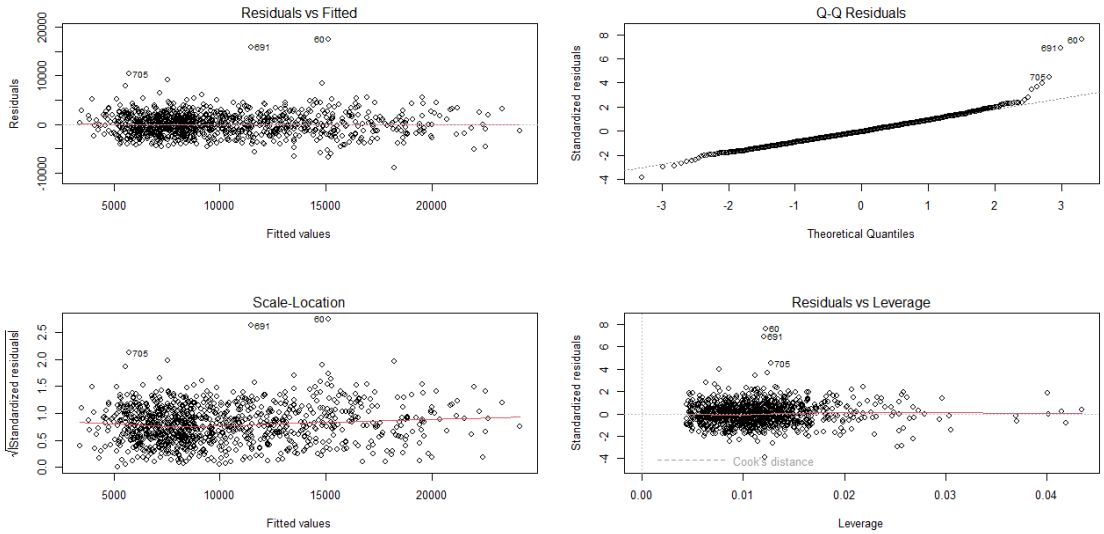
	Model 1a	Model 1b	Model 2a	Model 2b
Intercept	5181.38***	12063.86***	4033.68***	10359.93***
	(588.87)	(554.10)	(563.83)	(559.57)
Age (Years)	74.53***		68.49***	
	(6.31)		(5.93)	
BMI (Continuous)	155.25***		159.82***	
	(17.02)		(15.84)	
Sex: Male	-119.83	-1.34	-148.57	-43.07
	(155.59)	(163.98)	(145.12)	(152.58)
Current Smoker	7624.89***	7671.75***	7468.04***	7494.02***
	(201.34)	(212.83)	(188.85)	(199.19)
No Chronic Condition	-4100.75***	-4254.08***	-3801.33***	-3918.56***
	(182.84)	(192.72)	(173.27)	(183.66)
Exercise: Medium	-910.50***	-936.72***	-950.27***	-969.37***
	(182.60)	(192.95)	(169.94)	(179.11)
exercise_levelHigh	-1522.13***	-1589.43***	-1422.78***	-1476.31***
	(232.89)	(245.96)	(216.92)	(228.57)
Age: Senior		1917.78***		1747.02***
		(392.36)		(365.41)
bmi_catNormal Weight		-331.56		-81.56
		(508.89)		(472.85)
BMI: Overweight		327.21		592.39
		(500.11)		(464.81)

	Model 1a	Model 1b	Model 2a	Model 2b
BMI: Obese		1675.78**		1966.08***
		(509.16)		(473.20)
Prior Accidents			1013.33***	1172.74***
			(151.17)	(157.97)
Prior Claims			883.59***	822.65***
			(140.23)	(147.31)
Plan: Premium			1549.58***	1697.86***
			(214.65)	(225.44)
Plan: Standard			631.69***	718.32***
			(166.19)	(174.55)
Annual Checkups				27.43
				(66.66)
Num.Obs.	1031	1031	1031	1031
R2	0.709	0.678	0.750	0.725

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 3: Standardized Regression Coefficients (Ranked by Importance)

Variable	standerdised beta	Absolute standerdised beta
smokerSmoker	0.633	0.633
chronic_conditionNo Condition	-0.352	0.352
age	0.184	0.184
bmi	0.160	0.160
plan_typePremium	0.131	0.131
exercise_levelHigh	-0.122	0.122
prior_accidents	0.110	0.110
exercise_levelMedium	-0.103	0.103
prior_claims	0.103	0.103
plan_typeStandard	0.069	0.069
sexMale	-0.016	0.016


Residuals vs Fitted
Residuals
Fitted values
Q-Q Residuals
Standardized residuals
Theoretical Quantiles
Scale-Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage

Characteristic	**N = 1,031**[1]
sex	
Female	509 (49%)
Male	522 (51%)
smoker	
Non-Smoker	838 (81%)
Smoker	193 (19%)
region	
East	240 (23%)
North	242 (23%)
South	301 (29%)
West	248 (24%)
bmi_cat	
Underweight	30 (2.9%)
Normal Weight	271 (26%)
Overweight	434 (42%)
Obese	296 (29%)
age_cat	
Adult	984 (95%)
Senior	47 (4.6%)
chronic_condition	
Has Condition	247 (24%)
No Condition	784 (76%)
exercise_level	
Low	290 (28%)
Medium	542 (53%)
High	199 (19%)
plan_type	
Basic	315 (31%)
Premium	192 (19%)
Standard	524 (51%)

[1] n (%)